

Roots of Complex Numbers

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Based on the lecture note uploaded on SPeCTRUM

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Learning Outcomes

At the end of this lecture, you should be able to:

- Find the n^{th} roots of unity. $\leftarrow z^n = 1$
- Find the n^{th} roots of a complex number. $z^n = w$

n -th Roots

A complex number z is called an n^{th} root of another complex number w if $z^n = w$.

An n^{th} roots of unity is a complex number z satisfying the equation $z^n = 1$. $\leftarrow w=1$

Example: $1, i, -1, -i$ are the forth roots of unity

$$z^4 = 1 \Rightarrow z^4 - 1 = 0$$

$$(z^2 + 1)(z^2 - 1) = 0$$

$$[z^2 - (-1)](z+1)(z-1) = 0$$

$$(z+i)(z-i)(z+1)(z-1) = 0$$

$$z = -i, i, -1, 1$$

n -th Roots

To solve $z^n = w$ where n is a positive integer.

① Express w in polar form: $w = R(\cos \alpha + i \sin \alpha)$

② Let $z = r(\cos \theta + i \sin \theta)$, Then $z^n = w$ becomes

de Moivre's
then

$$r^n(\cos \theta + i \sin \theta)^n = R(\cos \alpha + i \sin \alpha)$$

$$r^n(\cos n\theta + i \sin n\theta) = R(\cos \alpha + i \sin \alpha)$$

cos & sin have period of 2π
 $n\theta = \alpha + 2k\pi, k \in \mathbb{Z}$

So, $r^n = R$ and $\theta = \frac{\alpha + 2k\pi}{n}, k \in \mathbb{Z}$. ($n\theta = \alpha + 2k\pi$)

The n^{th} roots of w are given by

substitute $k = 0, 1, \dots, n-1$

$$z_k = \sqrt[n]{R} \left[\cos \left(\frac{\alpha + 2k\pi}{n} \right) + i \sin \left(\frac{\alpha + 2k\pi}{n} \right) \right], \quad k = 0, 1, \dots, n-1$$

Remark:

$z^n = w$ has n distinct complex roots (with $k = 0, 1, \dots, n-1$).

n -th Roots

Example: Find the square roots of $2 + 2\sqrt{3}i$.

(m1)

$$\text{Let } z^2 = w = 2 + 2\sqrt{3}i$$

$$|w| = 4, \quad \text{Arg}(w) = \frac{\pi}{3} \quad (\text{Look at the last example in 3c})$$

$$\text{Let } z = r(\cos \theta + i \sin \theta)$$

$$z^2 = r^2(\cos(2\theta) + i \sin(2\theta)) = 4(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$$

$$\text{So, } r^2 = 4, \quad 2\theta = \frac{\pi}{3} + 2k\pi, \quad k \in \mathbb{Z}$$

$$r = 2 \quad (r \geq 0) \quad \theta = \frac{1}{2}(\frac{\pi}{3} + 2k\pi), \quad k \in \mathbb{Z}.$$

$$z_k = 2(\cos[\frac{1}{2}(\frac{\pi}{3} + 2k\pi)] + i \sin[\frac{1}{2}(\frac{\pi}{3} + 2k\pi)]), \quad k \in \mathbb{Z}.$$

$$k=0, \quad z_0 = 2(\cos(\frac{\pi}{6}) + i \sin(\frac{\pi}{6})) \quad \checkmark$$

$$k=-1, \quad z_{-1} = 2(\cos(\frac{7\pi}{6}) + i \sin(\frac{7\pi}{6})) \quad \checkmark$$

$$k=2, \quad z_2 = 2 \left[\cos \left(\frac{\pi}{6} + 2\pi i \right) + i \sin \left(\frac{\pi}{6} + 2\pi i \right) \right]$$

$$= 2 \left[\cos \left(\frac{\pi}{6} \right) + i \sin \left(\frac{\pi}{6} \right) \right] = z_0$$

n -th Roots

Example: Find the square roots of $2 + 2\sqrt{3}i$.

(m) Using Cartesian Form.

Let $z = x + yi$, $x, y \in \mathbb{R}$.

$$z^2 = 2 + 2\sqrt{3}i$$

$$(x + yi)^2 = 2 + 2\sqrt{3}i$$

$$x^2 + 2xyi - y^2 = 2 + 2\sqrt{3}i$$

$$x^2 - y^2 + 2xyi = 2 + 2\sqrt{3}i$$

Comparing real and imaginary part,

$$x^2 - y^2 = 2 \quad \text{--- (1)} \quad 2xy = 2\sqrt{3} \Rightarrow y = \frac{\sqrt{3}}{x} \quad \text{--- (2)}$$

$$\textcircled{1} \rightarrow \textcircled{2}, \quad x^2 - \left(\frac{\sqrt{3}}{x}\right)^2 = 2 \Rightarrow x^2 - \frac{3}{x^2} = 2$$

$$x^4 - 2x^2 - 3 = 0$$

$$(x^2 + 1)(x^2 - 3) = 0$$

$$x^2 = -1, 3$$

Since $x \in \mathbb{R}$, we take $x^2 = 3 \Rightarrow x = \pm\sqrt{3}$.

from ②, when $x = -\sqrt{3}$, $y = -1$

when $x = \sqrt{3}$, $y = 1$

So, $z = \sqrt{3} + i$ or $z = -\sqrt{3} - i$

n -th Roots

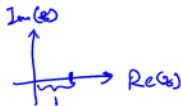
Example: Find the n^{th} roots of 1 for

(i) $n = 3$ ^{third roots of unity}

(ii) $n = 6$. ^{sixth roots of unity}

$$z^n = 1$$

$$= 1 (\cos 0 + i \sin 0)$$



$$\text{Let } z = r (\cos \theta + i \sin \theta)$$

$$z^n = r^n (\cos \theta + i \sin \theta)^n = 1 (\cos 0 + i \sin 0)$$

$$= r^n (\cos n\theta + i \sin n\theta) = 1 (\cos 0 + i \sin 0)$$

$$\text{Therefore, } r^n = 1 \quad n\theta = 0 + 2k\pi, \quad k = 0, 1, 2, 3, \dots$$

$$r = 1 \quad \theta = \frac{2k\pi}{n}$$

$$i) \quad n=3, \quad r=1, \quad \theta = \frac{2k\pi}{3}$$

$$k=0, 1, 2 \quad \nwarrow \quad n-1$$

$$k=0, \quad z_0 = 1 (\cos 0 + i \sin 0) = 1$$

$$k=1, \quad z_1 = 1 (\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}) = -\frac{1}{2} + \frac{\sqrt{3}}{2} i$$

$$k=2, \quad z_2 = 1 (\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}) = -\frac{1}{2} - \frac{\sqrt{3}}{2} i$$

$$k=3, \quad z_3 = 1 (\cos 2\pi + i \sin 2\pi) = 1 = z_0$$

$$ii) \quad n=6$$

$$\text{Exercise.} \quad z_0 \rightarrow z_5$$

n -th Roots

Example: Find the 6th roots of $w = 1 + \sqrt{3}i$.

Exercise.

$$z_0 = \sqrt[6]{2} \left(\cos \frac{9}{18} + i \sin \frac{9}{18} \right) \quad \downarrow \frac{6 \cdot 9}{18} = \frac{9}{3}$$

$$z_1 = \sqrt[6]{2} \left(\cos \frac{79}{18} + i \sin \frac{79}{18} \right) \quad \downarrow \frac{6 \cdot 9}{18}$$

$$z_2 = \sqrt[6]{2} \left(\cos \frac{129}{18} + i \sin \frac{129}{18} \right) \quad \downarrow \frac{6 \cdot 9}{18}$$

$$z_3 = \sqrt[6]{2} \left(\cos \frac{199}{18} + i \sin \frac{199}{18} \right) \quad \downarrow \frac{6 \cdot 9}{18}$$

$$z_4 = \sqrt[6]{2} \left(\cos \frac{259}{18} + i \sin \frac{259}{18} \right) \quad \downarrow \frac{6 \cdot 9}{18}$$

$$z_5 = \sqrt[6]{2} \left(\cos \frac{319}{18} + i \sin \frac{319}{18} \right) \quad \downarrow \frac{6 \cdot 9}{18}$$

$$z_6 = \sqrt[6]{2} \left(\cos \frac{379}{18} + i \sin \frac{379}{18} \right)$$

n -th Roots

Example: Solve the equation $z^4 = -8i$ and indicate your solutions in the Argand plane.

Try to do it in exponential form.

$$\text{Let } w = -8i$$

$$|w| = 8$$

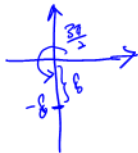
$$\arg(w) = \frac{3\pi}{2}$$

$$w = 8e^{i\frac{3\pi}{2}}$$

$$\text{Let } z = re^{i\theta}$$

$$z^4 = (re^{i\theta})^4 = 8e^{i\frac{3\pi}{2}}$$

$$r^4 e^{i4\theta} = 8e^{i\frac{3\pi}{2}}$$



$$r^4 = 8, \quad 4\theta = \frac{3\pi}{2} + 2k\pi$$

$$r = \sqrt[4]{8}, \quad \theta = \frac{3\pi}{8} + \frac{k\pi}{2}, \quad k=0,1,2,3$$

$$k=0, z_0 = \sqrt[4]{8} e^{i\frac{3\pi}{8}} \quad \text{---}$$

$$k=1, z_1 = \sqrt[4]{8} e^{i\frac{7\pi}{8}} \quad \text{---}$$

$$k=2, z_2 = \sqrt[4]{8} e^{i\frac{11\pi}{8}} \quad \text{---}$$

$$k=3, z_3 = \sqrt[4]{8} e^{i\frac{15\pi}{8}} \quad \text{---}$$

